MIL-T-26310A(USAF)
8 July 1959
Superseding
MIL-T-26310 (USAF)
29 April 1958

MILITARY SPECIFICATION

TRAILER, LIFT, AIRCRAFT - MISSILE ENGINE ETU-9/E

1. SCOPE

1.1 This specification covers an aircraft and missile engine lift trailer, designated ETU-9/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

TT-N-95	Naphtha; Aliphatic
GGG-W-641	Wrench, Bolt and Nut; Socket, Hand (Including Handles and Attachments)
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof--For Packaging and Sealing

Military

JAN-P-100	Packaging and Packing For Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Flexible
MIL-T-704	Treatment and Painting Of Material
MIL-C-3774	Crates, Open, Wood (2,500 to 10,000 Lb)
MIL-E-4970	Environmental Testing, Ground Support Equipment, General Specification For
MIL-T-5021	Tests; Aircraft Welding Operators' Certification
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance
MIL-H-6083	Hydraulic Fluid, Petroleum Base Preservative
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Specification For
MIL-A-8421	Air Transportability Requirements, General Specification For

FSC 1730

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

STANDARD

Military

MIL-STD-129 Marking For Shipment and Storage

PUBLICATION

Air Force-Navy Aeronautical Bulletin

143 Specifications and Standards; Use of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Consolidated Classification Committee

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 PREPRODUCTION.- This specification makes provisions for preproduction testing.

3.2 COMPONENTS.- The trailer shall consist of the following major components:

Description	See Requirements
a. Parallel rails	3.6
b. Hydraulic lift	3.7
c. Main Base	3.8
d. Winch	3.9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

3.3 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Bulletin 143 except as specified in 3.3.1 and 3.3.2.

3.3.1 COMMERCIAL PARTS.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 STANDARD PARTS.- With the exception specified in 3.3.1, AN and MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 DESIGN AND CONSTRUCTION.- The trailer shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 MAINTAINABILITY.- The trailer shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases without the use of special tools.

3.4.2 SAFETY FACTOR.- The trailer shall be designed for a safety factor of 3 based on the yield strength of the material.

3.4.3 MOBILITY.- The trailer shall be designed and constructed to meet the general requirements of MIL-M-8090 and those specified for type II mobility with the following additional requirements and limitations:

3.4.3.1 TIRES.- The tires shall be size 7.50 by 10, ten-ply pneumatic, or larger.

3.4.3.2 WHEELS

3.4.3.2.1 The four wheels shall be retractable by hydraulic lifts, with two wheels interconnected and steerable.

3.4.3.2.2 The four wheels shall turn 90° from the forward position and lock for side movement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

3.4.3.2.3 WHEEL TREAD.- The wheel tread shall be 72 ± 1 in.

3.4.3.2.4 WHEELBASE.- The wheelbase shall be a minimum of 148 in.

3.4.3.3 REFLECTORS.- Reflectors will not be required.

3.4.3.4 AIR TRANSPORTABILITY.- Air transportability requirements apply to the unloaded trailer.

3.4.3.5 TOWBAR.- A telescoping towbar 44-1/2 to 72 in. long shall be provided.

3.4.3.6 All towing operations shall be performed with the base frame at maximum ground clearance and with the lift frame fully collapsed. All locks shall be engaged.

3.4.4 LIFE EXPECTANCY.- The trailer shall be designed for a life expectancy of 10 yrs.

3.4.5 WELDERS.- Welding shall be accomplished by operators certified in accordance with MIL-T-5021, class A.

3.5 PERFORMANCE.- The trailer shall be capable of operating satisfactorily under the following conditions:

a. Temperatures ranging down to -65° F

b. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost

c. Exposure to salt-sea atmosphere

d. Fungus growth as encountered in tropical climates.

3.5.1 The trailer shall be capable of transporting a 10,000-lb component.

3.5.2 PROOFLOAD.- The trailer shall be capable of withstanding a proofload of 20,000 lbs. The center of gravity of the load shall be 29 in. above the center of the parallel rails and applied to two points on each rail 120 ± 2 in. apart.

3.5.3 The hydraulic system and winch shall be capable of withstanding repeated operations as specified in 4.5.6.1 and 4.5.6.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

3.6 PARALLEL RAILS.- The trailer shall have two parallel rails that shall be 174 ± 1/4 in. long. The rail gage shall be 48 ± 0.04 in. at the rail attachment and 48 ± 0.12 in. along the entire length of the rail. The rail shall have a 3.33 ± 0.125 in. working face on each rail. The working face or flange of each rail shall have a minimum clearance of 0.90 in. on the inside and a minimum clearance of 0.55 in. on the outside. These clearances shall extend 3 in. below the face of the rail. (See figure 1.) The end of each rail shall be square within +30, -0 minutes to insure contact on the top of the rails. The ends of the parallel rails shall be in the same plane within 1/16 in. The top face of the rails shall be in the same plane within 1/16 in.

3.6.1 ADJUSTMENTS.- The parallel rails shall be joined together to obtain the following minimum adjustments as shown on figure 1:

a. Yaw 1-3/4° in each direction from the centerline

b. Roll 10° each side of the horizontal centerline

c. Lateral 3 in. in each direction from the vertical centerline of the trailer

d. Ten degrees tilt from the horizontal at each end of the rails.

3.6.1.1 The mechanism for yaw, roll, and lateral adjustments shall be of the mechanical-type drive. The tilt adjustment shall be obtained by actuating the main hydraulic lift cylinders on each end of the trailer. The adjusting mechanisms shall be operable by hand while the trailer is under a rated load of 10,000 lbs. A minimum clearance radius of 21.5 in. with spacer shall be provided between the rails and structure for mounting required components.

3.6.1.2 Removable spacers shall be provided between the rails and lift structure to permit the rails to be lowered 5 in.

3.6.1.3 Control knobs shall have a 1/2-in. square drive for use with a 1/2-in. male wrench. The dimensions of the socket shall conform to GGG-W-641.

3.6.2 OVERLOAD.- The trailer shall be so designed that the adjustments specified in 3.7.1 can be accomplished with standard 1/2-in. drive wrenches when the trailer is under a 20,000-lb load. While under a 20,000-lb load, the trailer shall not tip or overturn when a 1,000-lb side load is applied at the side of the rails and the rails are raised to their maximum height with spacers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

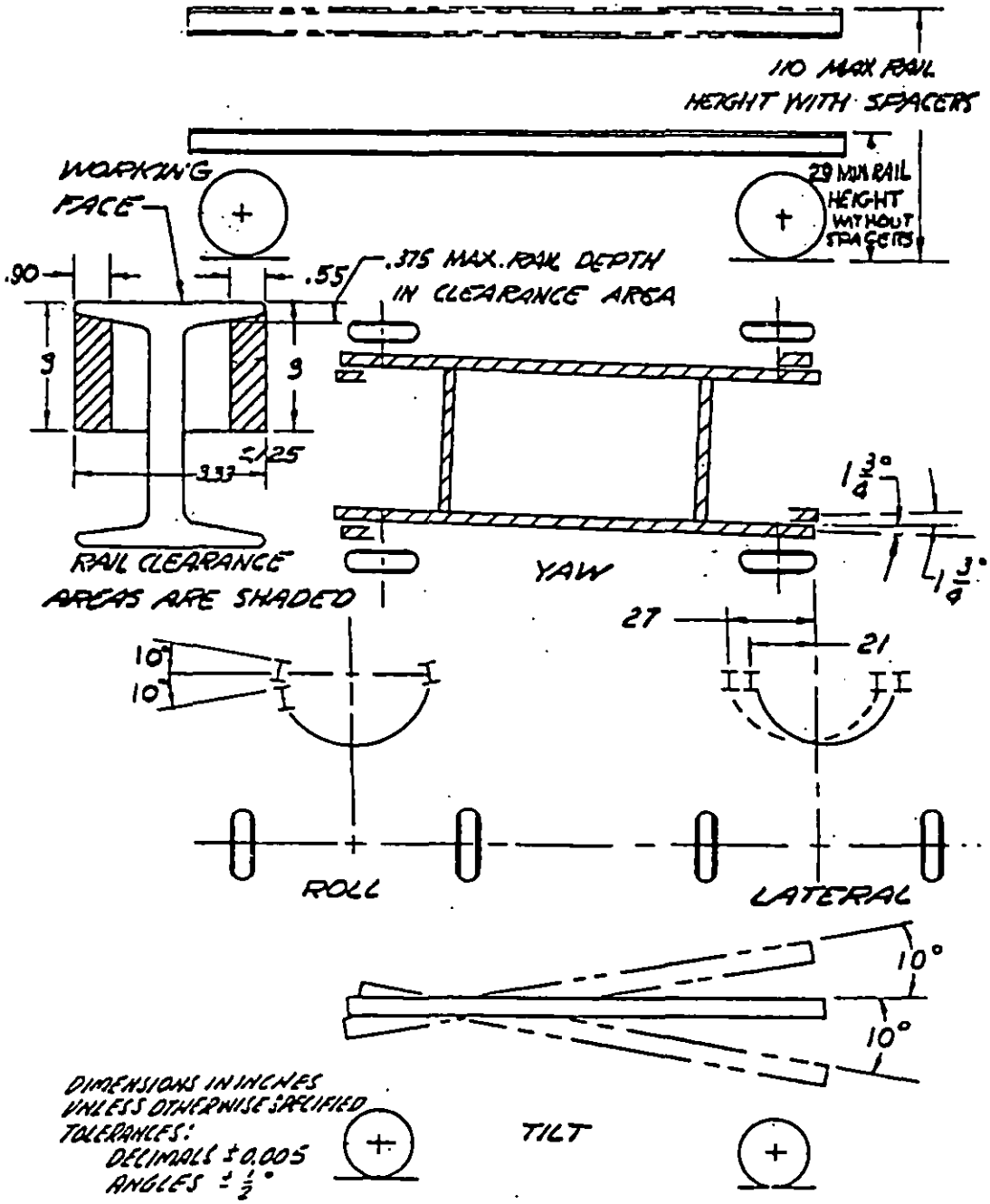


FIGURE 1 RAIL ADJUSTMENT & CLEARANCE

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

3.6.3 COUPLINGS.- Each rail shall be equipped with a male and female coupling. The mating dimensions of the couplings shall be in accordance with figure 2. The female coupling shall be provided with a quick-release, positive-lock-type pin. The pin shall be 3/4 in. in diameter, and permanently attached to the coupler by a steel cable. The pin shall have a double shear value in excess of 70,000 lbs.

3.6.4 RAIL GUIDES.- Each rail shall be provided with rail guides for aligning two rails together when two trailers are connected together. The rail guides shall be in accordance with figures 3 and 4.

3.6.5 RAIL STOPS.- A positive, mechanical-type stop shall be provided near the extreme ends of each rail. The stops shall be designed to hold a 10,000-lb component from rolling off the rails in the event of failure to lock a roller adapter to the rails. The stops shall be spring-loaded and manually retractable to permit a roller adapter to pass or roll along the rails to another trailer or into the aircraft. The rail stops shall be hand-operated and shall extend 1/2 in. above the face of the rails when fully extended. The rail stops shall engage the vertical face of the roller adapter.

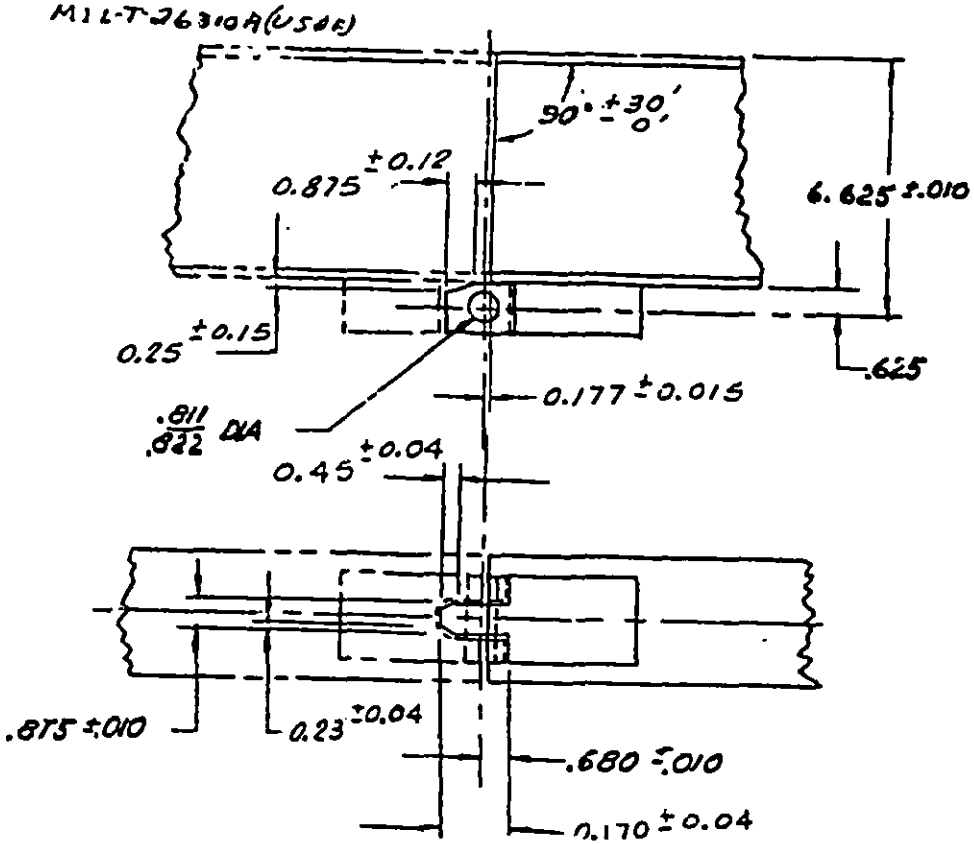
3.7 HYDRAULIC LIFT.- The hydraulic lift shall be provided with hydraulic cylinders. Each cylinder shall be equipped with a position-locking device to prevent lowering or tipping in the event of hydraulic line failure. The hydraulic lift shall raise the rails in a vertical direction with a variance of not more than 1/2 in. on either side of the true vertical.

3.7.1 The lift mechanism shall be designed to permit the center of gravity of the rated load to be transferred to either end (60 in. either side of center) and to prevent tipping. When the trailer is loaded to 10,000 lbs, the hydraulic system settling rate shall not exceed 0.02 in. per hour per cylinder. The settling rate shall be the same when the trailer is loaded to 20,000 lbs.

3.7.2 The lift structure shall be capable of adjusting the height of the rails from a low position, without spacers, of 29 in. to a maximum height of 110 in. with spacers under the rails. (See figure 1.) The trailer shall be designed to raise or lower the loads to any position within these limits.

3.7.3 LIFT CONTROLS.- The hydraulic lift provisions for the rails and wheels shall be located at one operator's position. They shall be designed to enable one operator to raise the rails or the main frame, maintaining a level attitude of the rails and frame. The provisions shall also enable one operator to control one end of the rail or frame.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCES: DECIMALS ±0.005

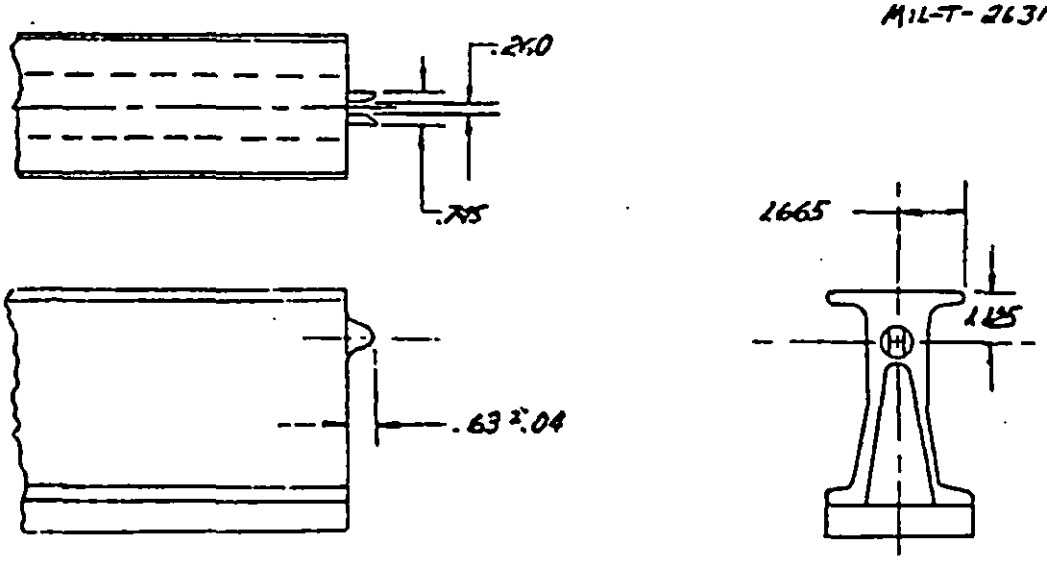
DIMENSIONS ARE SHOWN FOR MALE COUPLER ONLY

DIMENSIONS OF FEMALE COUPLER TO SUIT MALE COUPLER

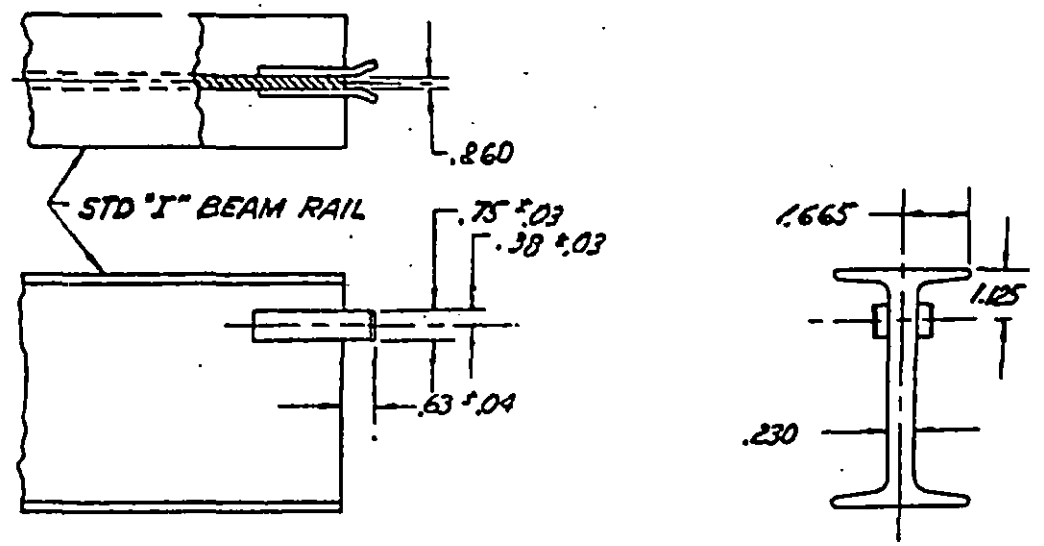
LOCATE MALE COUPLER ON RIGHT & FEMALE ON LEFT WHEN FACING RAIL ENDS

FIGURE 2 ENVELOPE & MATING DIMENSIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.



HEAVY WEB



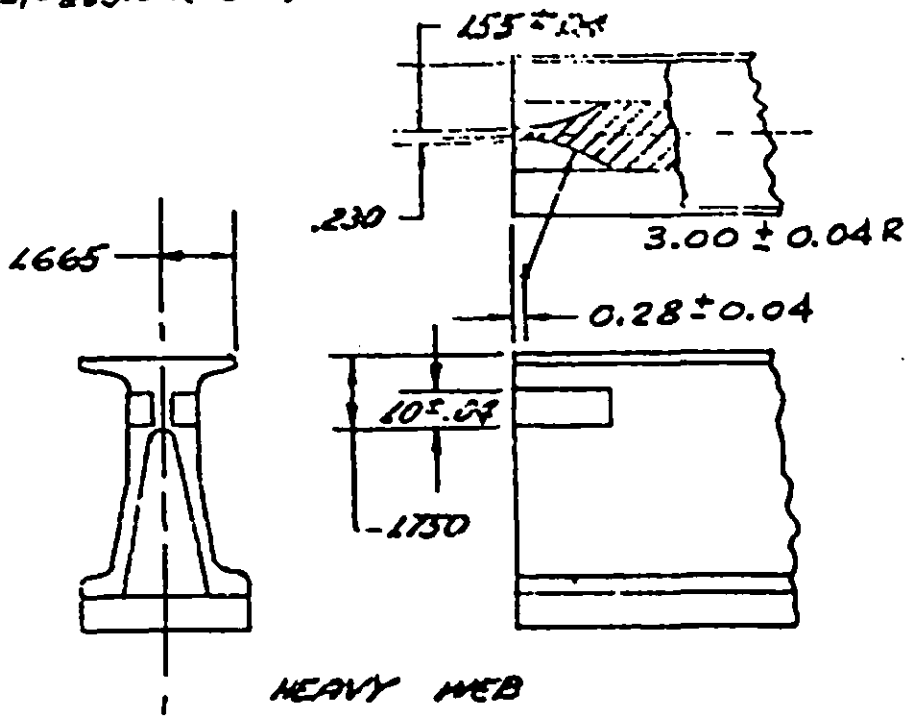
LIGHT WEB

WHEN FACING RAIL ENDS THE RIGHT HAND RAIL GUIDE SHALL MATE WITH BOTH GUIDES SHOWN ABOVE

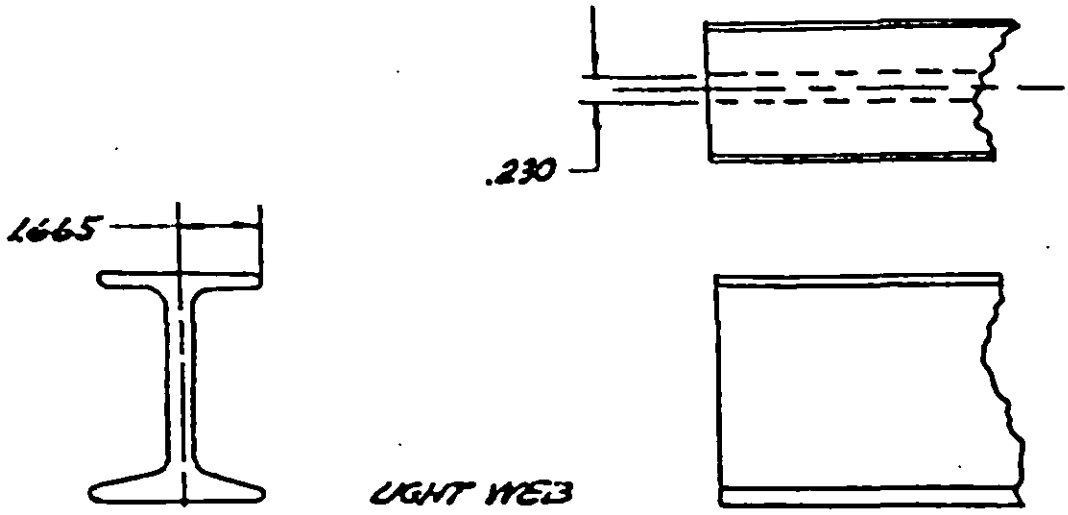
DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED TOLERANCE ± 0.015

FIGURE 3 RAIL GUIDE ENVELOPE RIGHT HAND

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.



HEAVY WEB



LIGHT WEB

WHEN FACING RAIL ENDS THE LEFT HAND RAIL GUIDE SHALL MATE WITH BOTH GUIDES SHOWN ABOVE

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCE: DECIMALS ± 0.015

FIGURE 4. RAIL GUIDE ENVELOPE LEFT HAND

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

3.8 MAIN BASE.- The main base supporting the structure shall be capable of holding the load in any position. Four leveling posts shall be provided on the main base that are independently operated and shall have at least 12 in. of adjustment. A means shall be provided to hydraulically raise the main base independently at each end by raising or lowering the wheels. The hydraulic lifts shall be equipped with positive locks in case of hydraulic failure. The lifts shall be capable of raising or lowering a 10,000-lb load.

3.9 WINCH.- A hand-operated, positive-control winch shall be provided on the lift frame on a center with the parallel rails. The winch shall be geared to enable a 10,000-lb load to be moved on the rails with four rollers by hand cranking when the rails are tilted 10° from the horizontal (fore and aft). The maximum load on the handle shall not exceed 26 lbs. The winch shall operate in either a compression or tension loading direction.

3.9.1 ATTACHMENT BAR.- The winch shall have a telescoping attachment bar that is adjustable from 35 to 60 in. for attachment to the engine or component and for driving the component along the rails. The free end of the attachment bar shall be provided with a terminal fitting in accordance with figure 5. A quick, positive means shall be provided for changing the attachment bar for different attachment points on the engine or component or reversed to the opposite end to permit movement on the rails.

3.9.2 TRACK.- The winch shall be provided with a track or travel media of a minimum length of 112 in. total travel.

3.9.3 OVERLOAD PROTECTION DEVICE.- The winch shall be provided with an overload protection device to prevent damage to the aircraft or component while the component is being installed or removed. The overload protection device shall be set to release at 1,600 to 1,700 lbs tension.

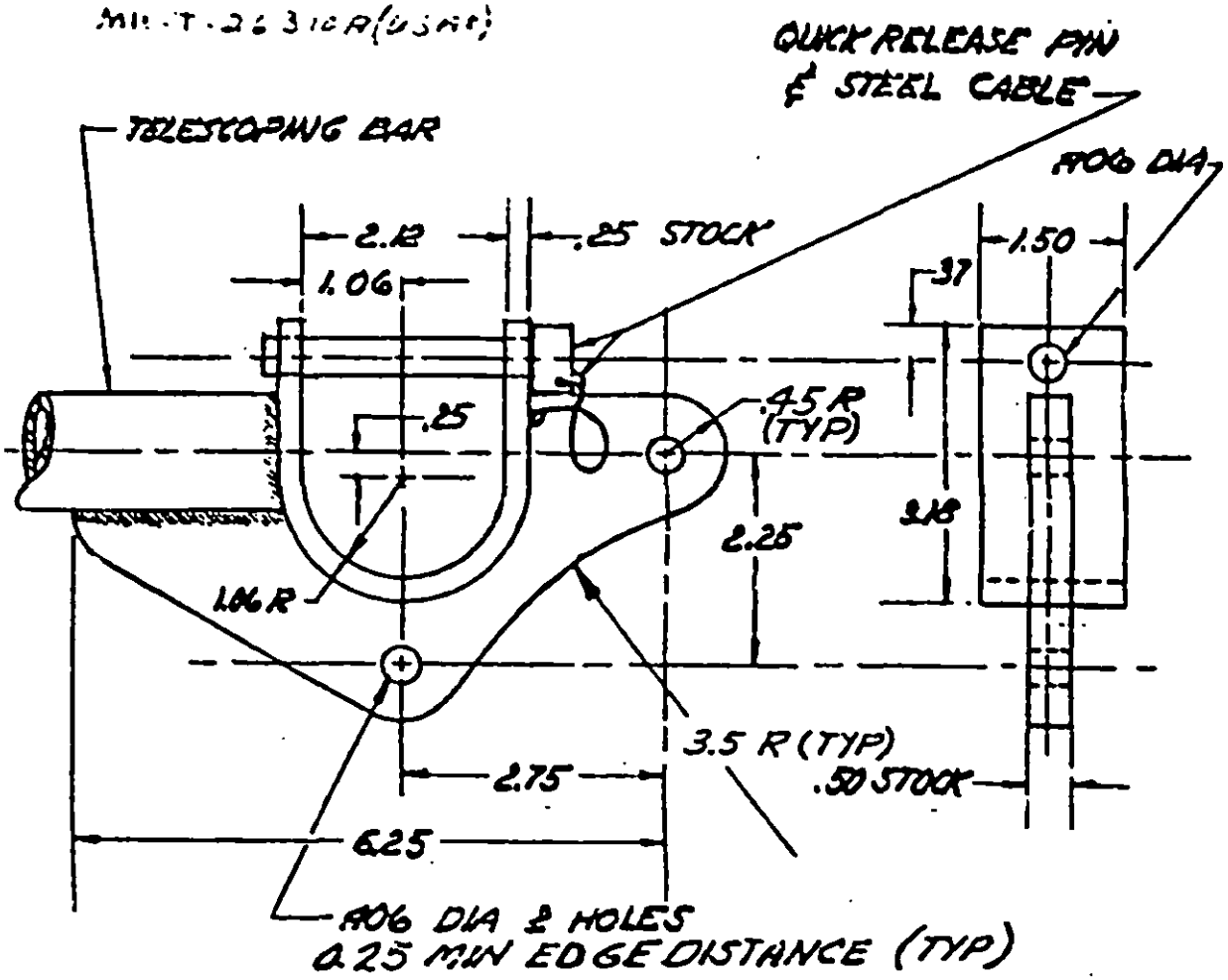
3.10 LUBRICATION.- Where practicable, permanently lubricated components or units not requiring lubrication shall be used. Installation of grease and oil seals shall provide accessibility for replacement and, where feasible, bearing surfaces shall be renewable.

3.11 WEIGHT.- The weight of the trailer shall not exceed 3,500 lbs.

3.12 FINISH.- The trailer shall be finished in accordance with MIL-T-704, type A.

3.13 WARNING PLATES.- Permanently attached warning plates shall be provided in prominent positions on the trailer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.



DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES ±0.010

FIGURE 5 TERMINAL FITTING

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

3.13.1 A permanent-type caution plate containing the following information shall be securely attached to both sides of the trailer body:

DON'T USE FORKLIFTS OR STACK
THIS TRAILER WHEN UNCRATED.

3.14 WORKMANSHIP.- The trailer, including all parts and accessories, shall be constructed and finished in a workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies, welding and brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.14.1 RIVETING.- Riveting operations shall be performed to insure that the rivets are tight and satisfactorily headed.

3.14.2 CLEANING.- Prior to the application of the surface treatments, cleaning shall be accomplished as specified herein with materials and processes that have no deleterious effect on the metal and by means of a procedure that is satisfactory for the application of the surface treatment required.

3.14.2.1 The trailer shall be thoroughly cleaned of loose, spattered or excessive weld scale and flux, rust and metal chips, oil, grease, oxides, and other foreign matter after assembly. Burrs and sharp edges, as well as resin finish that may crumble, crack, et cetera, shall be removed.

3.14.2.2 MULTIPLE-PASS WELDS.- When a weld is built up by two or more beads or passes, preceding beads shall be cleaned free of scale and flux by chipping or scraping and brushing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of the trailer shall be classified as follows:

a. Acceptance tests. . . . . See 4.3

b. Preproduction testing . . . . . See 4.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

4.3 ACCEPTANCE TESTS.- The acceptance tests shall consist of the individual tests.

4.3.1 INDIVIDUAL TESTS.- Each trailer shall be subjected to the following tests, as described under 4.5:

a. Examination of product. . . . . See 4.5.1

b. Load test . . . . . See 4.5.2

4.4 TEST CONDITIONS

4.4.1 During the test specified in 4.5.5.1, the hydraulic system shall be serviced with a mixture by volume of 75 percent hydraulic fluid conforming to MIL-H-5606 and 25 percent aliphatic naphtha conforming to TT-N-95.

4.5 TEST METHODS

4.5.1 EXAMINATION OF PRODUCT.- The trailer shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, dimensions, and marking.

4.5.2 LOAD TEST.- A 10,000-lb load shall be placed in the center of the rails support from two points on each rail and raised. The load shall then be shifted to each end of the trailer and the suitability of the hydraulic cylinders and winch drive shall be proven. The load shall be applied in such a manner as to enable compression and tension loads to be applied to all lift cylinders and the winch drive. During this test, all manipulators shall be operated through their entire ranges. No binding or roughness of operation shall be evident.

4.5.3 TOWING.- While carrying a 10,000-lb load (in low position), the trailer shall be tested for type II mobility in accordance with MIL-M-8090. The low position shall be with the wheels in maximum height position and the rails in lowest position with the spacers in place.

4.5.4 STATIC LOAD TESTS.- The trailer shall be subjected to a static load of 20,000 lbs. The load shall be raised to maximum height with spacers and lowered to the low position while loaded. The lifts provided on the wheels shall also be subjected to this static load and operated through their entire range. No damage shall result from the static load. The lift cylinders shall exhibit no leaks around their seals and the hydraulic system shall be suitable for these entire loads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

4.5.4.1 SETTLING TESTS.- During the tests specified in 4.5.4, the settling rate of the hydraulic cylinders shall be measured. The cylinder settling rate shall not exceed 0.02 inch per hour per cylinder. The hydraulic locks will not be used during the settling test.

4.5.4.2 TENSION LOAD TEST.- The tension of the hydraulic cylinders shall be tested by placing 10,000 lbs on one end of the trailer. The loaded end of the rails shall not deflect more than 4 in. down and the opposite end of the rails shall not deflect more than 5 in. when measured from the top of the rail to the top of the main frame. This test shall be conducted in four positions with the lift being extended one-fourth of its maximum lift for each test. This test shall be repeated on the opposite end of the trailer in order to prove the adequacy of all lifting cylinders.

4.5.4.3 The tilt, roll, yaw, and transverse mechanisms shall be operated through their entire ranges and shall be tested under a 20,000-lb load. The adjusting mechanism shall be operable by a standard 1/2 in. drive wrench and shall not require more than 700 in. lbs of torque with the load in any position on the rails. The winch drive shall be demonstrated by rolling this load from each end of the rails. If the torque required to perform all adjustments exceeds 700 in. lbs, the trailer shall be rejected.

4.5.4.4 SIDE LOAD TESTS.- The trailer shall be raised to its maximum height. The trailer shall be subjected to a static load of 20,000 lbs. A side load of 1,000 lbs shall be applied at the side of the rails. The trailer shall not exhibit permanent deformation, tip, or overturn when the 1,000-lb side load is applied. The trailer shall not be tied down during this test.

4.5.4.5 RAIL STOP TESTS.- A 10,000-lb load shall be mounted on roller adapters and located at one end of the trailer rails. The trailer shall be tilted to its maximum tilt. The 10,000-lb load shall be permitted to roll free to the rail stop. The stop shall not fail or permit the load to roll from the trailer. The load shall roll from one-half the length of the rail. Failure of the rail stops shall be cause for rejection of the trailer.

4.5.4.6 FLIGHT AND TAXIING LOAD TESTS.- The trailer shall be attached to a pallet in the manner specified for loading in an aircraft. The trailer shall then be tested as specified in MIL-A-8421.

4.5.5 ENVIRONMENTAL TESTS.- The trailer or critical components thereof shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-E-4970. Upon

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

completion of the specified test period, the trailer or component shall be examined, tested, and operated under the applicable conditions specified herein.

4.5.5.1 LOW TEMPERATURE.- The trailer shall be subjected to the test specified in procedure I, except the trailer shall be subjected to -65° F only for a period of 72 hrs. At the end of the soaking period, the trailer shall be raised to its maximum height. The lowering valves shall be opened and the rams shall retract with not more than 75 lbs applied to the trailer. The time and number of full strokes necessary to fully extend all rams shall be recorded at +70° and -65° F. The time and strokes required to fully extend all rams at -65° F shall be not more than double the time and strokes required at +70° F. The trailer shall then be tested in accordance with 4.5.2.

4.5.5.1.1 Upon completion of the testing specified in 4.5.5.1, the loaded trailer shall be withdrawn from the chamber and immediately road tested over improved roads for 5 miles at 20 mph and over unimproved roads for 5 miles at 5 mph. No failures or signs of impending failure shall be evident at the conclusion of the test.

4.5.5.1.2 After the road test, the trailer shall be operated through all its operations under rated load. The lift mechanisms and the tilt, yaw, and roll and winch drive shall be operated through their full range. They shall not be affected by the low temperature or road tests.

4.5.5.2 HUMIDITY.- The trailer shall be subjected to humidity in accordance with the specified procedure.

4.5.5.3 SALT SPRAY.- The trailer shall be subjected to salt spray in accordance with the specified procedure for 50 hrs.

4.5.5.4 FUNGUS RESISTANCE.- The hydraulic lift mechanisms shall be subjected to fungus in accordance with the specified procedure.

4.5.6 ENDURANCE TESTS

4.5.6.1 HYDRAULIC SYSTEM.- An endurance test shall be conducted on the hydraulic system of the trailer. The test shall consist of 50 cycles of raising 10,000 lbs. One cycle shall consist of raising the trailer from collapsed height to the fully extended height and return to the collapsed position. At the end of the endurance test, the force required at the end of the pump handle(s) to raise the 10,000 lbs shall not exceed 75 lbs each. After the endurance test, the cylinder settling rate shall not exceed 1/16 inch per hour. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

settling rate shall be measured at 10, 50, and 90 percent of height adjustment.

4.5.6.2 WINCH.- An endurance test shall be conducted on the winch. The test shall consist of 50 cycles of rolling a 10,000-lb load on the rails tilted 10° from the horizontal. One cycle shall consist of rolling the load from one end of the rails to the opposite end and return to the starting position. At the end of the endurance test, the maximum load on the handle shall not exceed 26 lbs. The rails shall be tilted 10° to enable a tension load for 25 cycles and a compression load for 25 cycles to be applied to the winch. During the endurance testing, the overload protection device shall demonstrate satisfactory operation in compliance with 3.9.

4.6 PREPRODUCTION TESTING

4.6.1 PREPRODUCTION TEST SAMPLE.- The preproduction test sample shall consist of one trailer representative of the production equipment. It shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract (see 6.2), at the contractor's plant under the supervision of the procuring activity.

4.6.2 PREPRODUCTION TESTS.- The preproduction tests shall consist of all the tests described under 4.4.

4.7 EXAMINATION AND TESTS OF PREPARATION FOR DELIVERY REQUIREMENTS.- The preservation, packaging, packing, and marking of the equipment shall be subjected to examination and tests to determine compliance with the requirements of section 5 of this specification. The contractor shall provide the necessary facilities and supplies for inspection and tests made prior to final acceptance.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVEL A.- The trailer shall be preserved and packaged in accordance with method I of MIL-P-116.

5.1.1.1 HYDRAULIC SYSTEM.- The hydraulic system shall be filled with hydraulic fluid conforming to MIL-H-6083. Openings shall be covered with grade A paper conforming to MIL-B-121 and sealed with water-resistant tape conforming to PPP-T-60. A waterproof tag clearly marked as follows shall be secured to the trailer:

CAUTION: Drain corrosion-preservative oil from the hydraulic system and refill with operational oil specified in technical order for the equipment before placing in operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

5.1.1.2 DISASSEMBLY AND PACKAGING.- The towbar and rails with space blocks shall be disassembled from the trailer. Nuts, bolts, and washers as a result of disassembly shall be secured to the larger assembly with which they connect. The disassembled parts shall be nested, blocked, braced, and cushioned in the forward and after tubes.

5.1.2 LEVEL C.- The trailer shall be afforded preservation and packaging that will insure sufficient protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.2 PACKING

5.2.1 LEVEL A.- The adjusting and positioning mechanism shall be adjusted to level and the trailer lowered to its most descended position. The wheels shall be fully retracted so that the load rests on the frame. The trailer, preserved and packaged as specified in 5.1.1, shall be securely fastened in a crate conforming to MIL-C-3774.

5.2.2 LEVEL B AND AIR SHIPMENT.- The trailer, preserved and packaged as specified in 5.1.1 and 5.1.1.2 shall be afforded sufficient protection against deterioration and damage during shipment from the supply source to the storage site. The protection provided shall insure acceptance by the carrier(s) at the lowest packing, car-loading, and freight costs combined, compatible with the quantity for shipment to a single destination and shall conform to the uniform freight classification rules and regulations of the carrier(s) applicable to the mode of transportation.

5.2.3 LEVEL C.- The trailer shall be afforded sufficient protection to insure against deterioration and damage during direct shipment from the supply source to the first receiving activity for immediate use. The protection provided shall insure acceptance by the carrier at the lowest packing, car-loading, and freight costs combined, compatible with the quantity for shipment to a single destination and shall conform to uniform freight classification rules and regulations applicable to the mode of transportation.

5.3 PHYSICAL PROTECTION.- Cushioning, blocking, bracing, and bolting as required shall be in accordance with JAN-P-100. The drop tests specified in JAN-P-100 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

5.4 MARKING.- Interior and exterior containers and trailers shipped devoid of container shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows:

TRAILER, LIFT, AIRCRAFT - MISSILE ENGINE ETU-9/E

6. NOTES

6.1 INTENDED USE.- The ETU-9/E trailer covered by this specification is intended for use in installing or removing aircraft engines and other components from aircraft and missiles and for the ground transportation and handling of these components. Also, it is intended for use in some forms of field maintenance of aircraft engines.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Point of testing (see 4.6.1)

c. Required levels of preservation and packaging, and packing (see section 5).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

KLM/BLW/mt/mr
WILE

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO If "YES", in what way?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD, FORM 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE PLATE NO. 15618

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T09:26Z
Check the source to verify that this is the current version before use.